Uplifting trimAl for handling thousands of sequences

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https://github.com/inab/trimal

Multiple Sequence Alignment

A multiple sequence alignment (known as **MSA**) refers to the alignment of three or more biological sequences. MSAs play a central role in many areas of bioinformatics, including homology modeling, database searches, structure prediction, genome annotation, motif finding and **phylogeny inference**. MSAs are crucial in all of these applications with great impact on the reliability and accuracy of downstream analyses.

The importance of the quality of alignments for downstream analyses, the generation of massive amounts of MSAs as part of nowadays common large-scale analysis and the possibility to generate **large-scale alignments** including tens to hundreds of thousands sequences underscore the need for better strategies to improve the quality of MSAs.

trimAl
A tool for automated alignment trimming

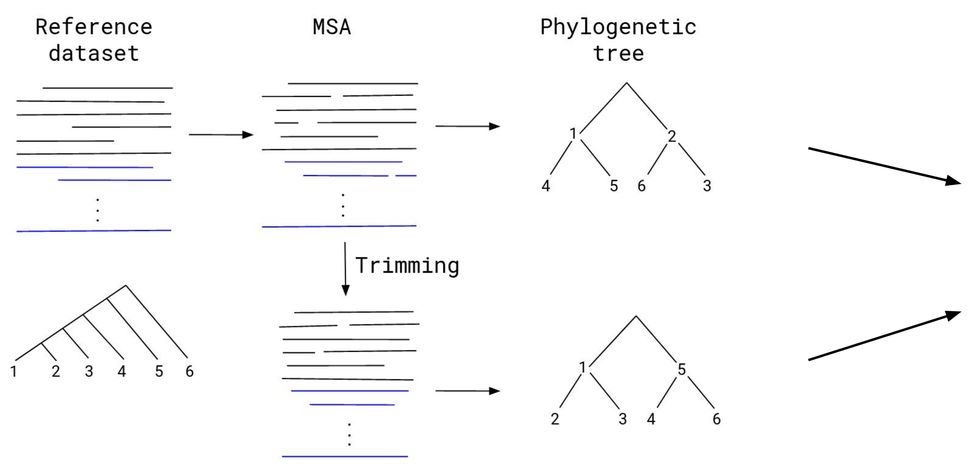
trimAl is a tool for automated removal of spurious sequences or poorly aligned regions from a multiple sequence alignment. It can consider several parameters in order to select the most-reliable positions in the alignment. The programme implements a series of **automated algorithms** that trim the alignment searching for optimum thresholds based on inherent characteristics of the input alignment, to be used so that the **signal-to-noise ratio** after alignment trimming phase is increased.

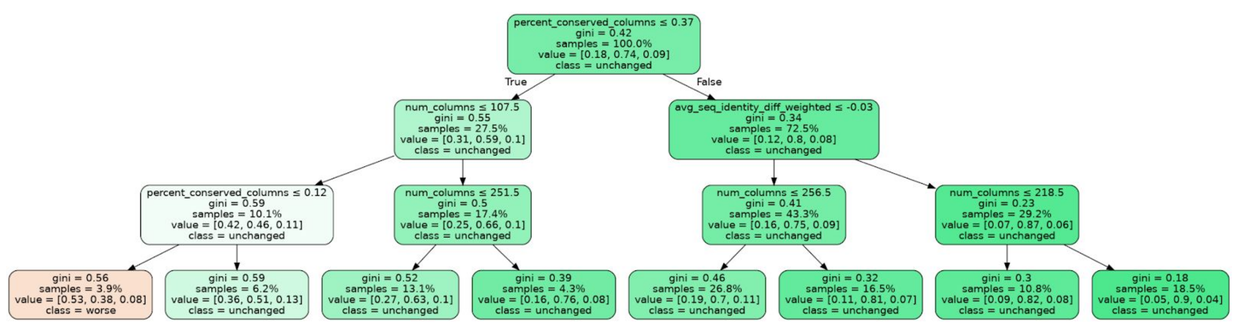
What have we done?

We show how the use of **machine-learning** techniques allows us to generate new **automated filtering algorithms** based on the most relevant MSA features. The dataset we used is composed of sequences of six orthologs of **undisputed reference topology** and it was enriched with several homologues. Besides, it contains information for many different MSAs and automated trimming programs, including a previous trimAl version.

This work is complemented by additional investigations on how to extend the **benchmark** to measure the impact of trimming on (very) **large alignments** in terms of global accuracy. Finally, we demonstrate the **technical capabilities** of the newest trimAl version when there are no limitations in terms of computational resources.

Machine learning model





New automated filtering algorithms

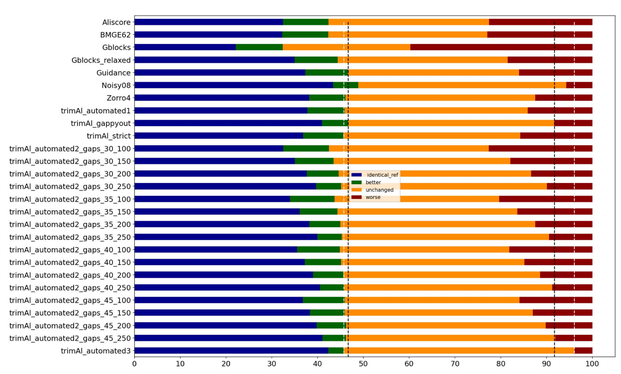
Automated2

1. Sort columns by gap score
2. Select a % of the MSA
3. Add columns if less than the minimum

Automated3

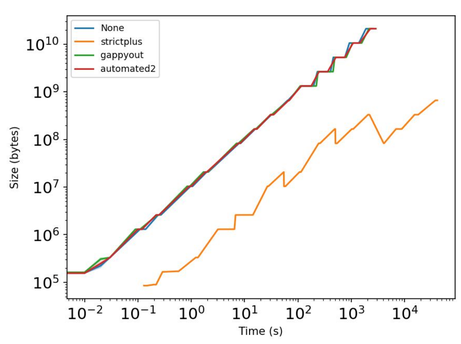
1. Apply trimAl's gappyout method
2. Add columns if less than the minimum

Unfiltered vs filtered MSAs



Benchmarking of massive MSAs

Technical capabilities



	Columns	Accuracy (sum-of-pairs)	% conserved columns
Original	334 ± 304	0.70 ± 0.19	
Gappyout	190 ± 148	0.86 ± 0.12	0.66 ± 0.23
Strictplus	149 ± 116	0.90 ± 0.09	0.53 ± 0.23
Automated2	215 ± 120	0.76 ± 0.18	0.81 ± 0.23

Conclusions

- The size of the original MSA and the % of conserved columns after trimming are the two main features affecting the quality of the reconstructed tree
- Trimming without affecting the downstream topology might have a significant impact on the use of computational resources
- Automated trimming has shown as a good option in terms of accuracy measured as the sum-of-pairs

Next steps

- Different datasets should be analyzed in order to confirm those findings (e.g. DNA MSAs)
- Further studies should consider generating datasets of undisputed phylogenetic with a larger number of ortholog sequences to provide a more realistic scenario

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